



THE KING'S SCHOOL

2009 Higher School Certificate Course Trial Examination

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1-10
- All questions are of equal value

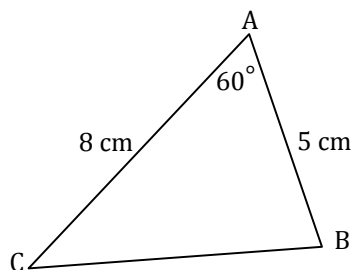
Total marks – 120
Attempt Questions 1-10
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks) Use a SEPARATE writing booklet.	Marks
(a) Find the value of $12e$ to 2 decimal places.	2
(b) Solve $7 - 8x < -9$	2
(c) Simplify $\frac{1 + x + x^{11} + x^{12}}{1 + x}$	2
(d) Find the perpendicular distance from the point $\left(\frac{1}{2}, 1\right)$ to the line $6x + 8y + 1 = 0$	2
(e) $f(x) = \frac{12 + x^2}{10}$ Find $f'(60)$	2
(f) Show on a number line the solutions of the inequality $x(2 - x) > 0$	2

End of Question 1

(a)



- | | | |
|------|--|----------|
| (i) | Find the exact area of $\triangle ABC$ | 2 |
| (ii) | Find the length of side BC | 2 |

(b) Differentiate

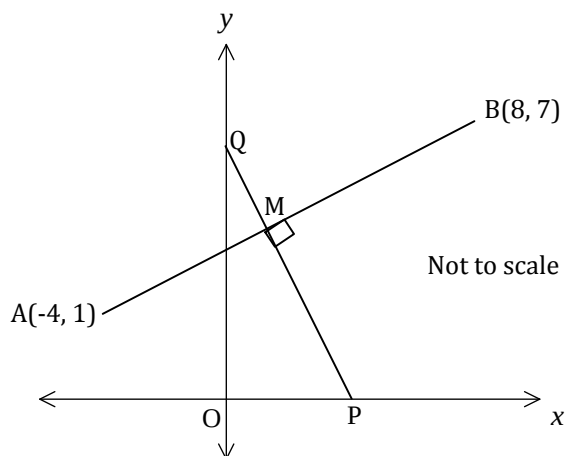
- | | | |
|------|--------------|----------|
| (i) | $(1 + 2x)^3$ | 1 |
| (ii) | $x \ln x$ | 2 |

(c) The curve $y = f(x)$ passes through the point $P(1, 3)$ where $f'(x) = 1 + 2x - 3x^2$

- | | | |
|-------|---|----------|
| (i) | Show that $P(1, 3)$ is a stationary point. | 1 |
| (ii) | Determine the precise type of stationary point at $P(1, 3)$. | 2 |
| (iii) | Find $f(x)$ | 2 |

End of Question 2

(a)



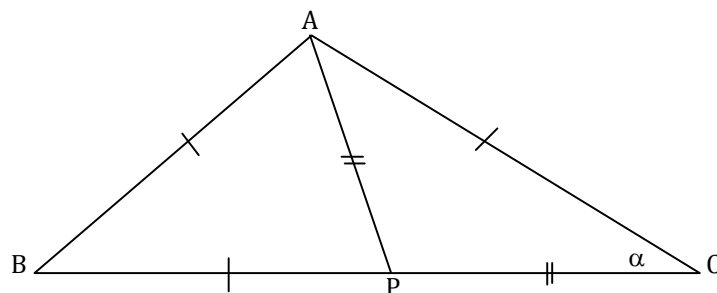
$A(-4, 1)$ and $B(8, 7)$ are two points in the number plane.

M is the midpoint of AB and the perpendicular bisector of AB meets the x axis at P and the y axis at Q.

- | | |
|---|----------|
| (i) Find the coordinates of M. | 1 |
| (ii) Show that the equation of the perpendicular bisector of AB, i.e. line QMP, is $2x + y = 8$ | 2 |
| (iii) Hence find the coordinates of the points P and Q. | 1 |
| (iv) Show that the quadrilateral APBQ is a rhombus. | 2 |
| (v) Hence, or otherwise, find the area of the rhombus. | 2 |

Question 3 continues on the next page

(b)



$\triangle ABC$ is isosceles where $AB = AC$

P is the point on BC where $AB = PB$ and $AP = PC$

Let $\angle ACP = \alpha$

(i) Prove that $\angle APB = 2\alpha$ 2

(ii) Find the value of α [YOU DO NOT NEED TO GIVE REASONS IN PART (ii)] 2

End of Question 3

-
- (a) Find a primitive function of $\frac{12x^3}{x^4 + 1}$ **2**
- (b) Use Simpson's Rule with 3 function values to find an approximate value of the
integral $\int_0^2 \sqrt{3x^3 + 1} \, dx$ **3**
- (c) (i) State the vertex of the parabola $(x - 1)^2 = -8(y + 2)$ **1**
- (ii) Find the equation of the directrix of this parabola. **2**
- (d) $x^2 - 4x + 2 = 0$ has the roots α, β
- (i) Write down the values of $\alpha + \beta$ and $\alpha\beta$ **2**
- (ii) Hence find a quadratic equation with the roots $3\alpha, 3\beta$ **2**

End of Question 4

-
- (a) Consider the series $30 + 21 + \dots$
- (i) If the series is arithmetic, find the sum of the first 14 terms. **2**
- (ii) If the series is geometric, find the limiting sum. **2**
- (b) Let $f(x) = 4\cos 2x$
- (i) State the period of the function. **1**
- (ii) Sketch the graph of the function for $0 \leq x \leq \pi$ **2**
- (iii) Find the area of the region in the first quadrant bounded by the curve and the coordinate axes. **3**
- (c) (i) Sketch the graph of $y = |x - 2|$ **1**
- (ii) Hence, or otherwise, evaluate $\int_0^4 |x - 2| dx$ **1**

End of Question 5

- (a) A particle moves on the x axis according to the displacement equation of motion

$$x = t + \frac{16}{t+1} - 8, \quad t \geq 0$$

t is time in seconds and x is measured in metres.

- | | |
|---|---|
| (i) Find the initial displacement. | 1 |
| (ii) Find the times when the particle stops. | 2 |
| (iii) Find the distance travelled by the particle during the first 7 seconds. | 2 |
| (iv) Find the maximum acceleration. | 2 |
- (b) The population P of a town is increasing exponentially so that
- $$\frac{dP}{dt} = kP, \quad k \text{ a positive constant, } t \text{ time in years}$$
- | | |
|---|---|
| (i) Show that $P = Ae^{kt}$ is a solution of $\frac{dP}{dt} = kP$, where A is a constant. | 1 |
| (ii) Initially the population was 4 000 and 12 years later it was 5 400.
Find the value of k correct to 2 significant figures. | 3 |
| (iii) What was the initial rate of increase of the population in people/year? | 1 |

End of Question 6

- (a) Joni borrows $\$P$ for an overseas holiday. This loan plus interest is to be repaid in equal monthly instalments of \$232 for 5 years.

Interest of 6% p.a. is charged monthly on the balance owing at the start of each month.

Let $\$A_n$ be the amount owing at the end of n months.

(i) Show that $A_1 = P(1.005) - 232$ **1**

(ii) Show that $A_2 = P(1.005)^2 - 232(1 + 1.005)$ **1**

- (iii) Write down a similar expression for A_n and deduce that

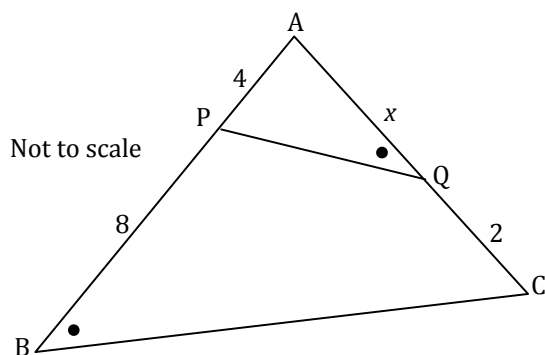
$$A_n = P(1.005)^n - 46\,400(1.005^n - 1)$$
 2

- (iv) Find, in whole dollars, the amount that Joni borrowed. **2**

- (v) How much will Joni owe after 1 year of monthly instalment payments? **1**

Question 7 continues on the next page

(b)



In the diagram, $AP = 4$, $PB = 8$, $QC = 2$ and $\angle ABC = \angle AQP$

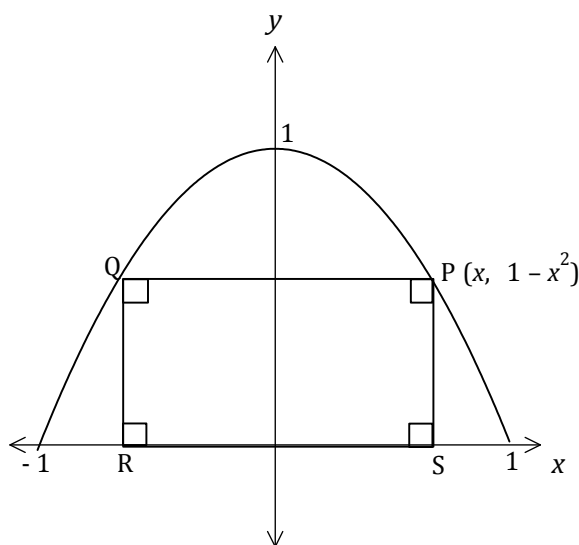
Let $AQ = x$

(i) Prove that $\triangle ABC$ is similar to $\triangle APQ$. 2

(ii) Find x . 3

End of Question 7

(a)



The sketch shows the parabolic arc $y = 1 - x^2$, $-1 \leq x \leq 1$

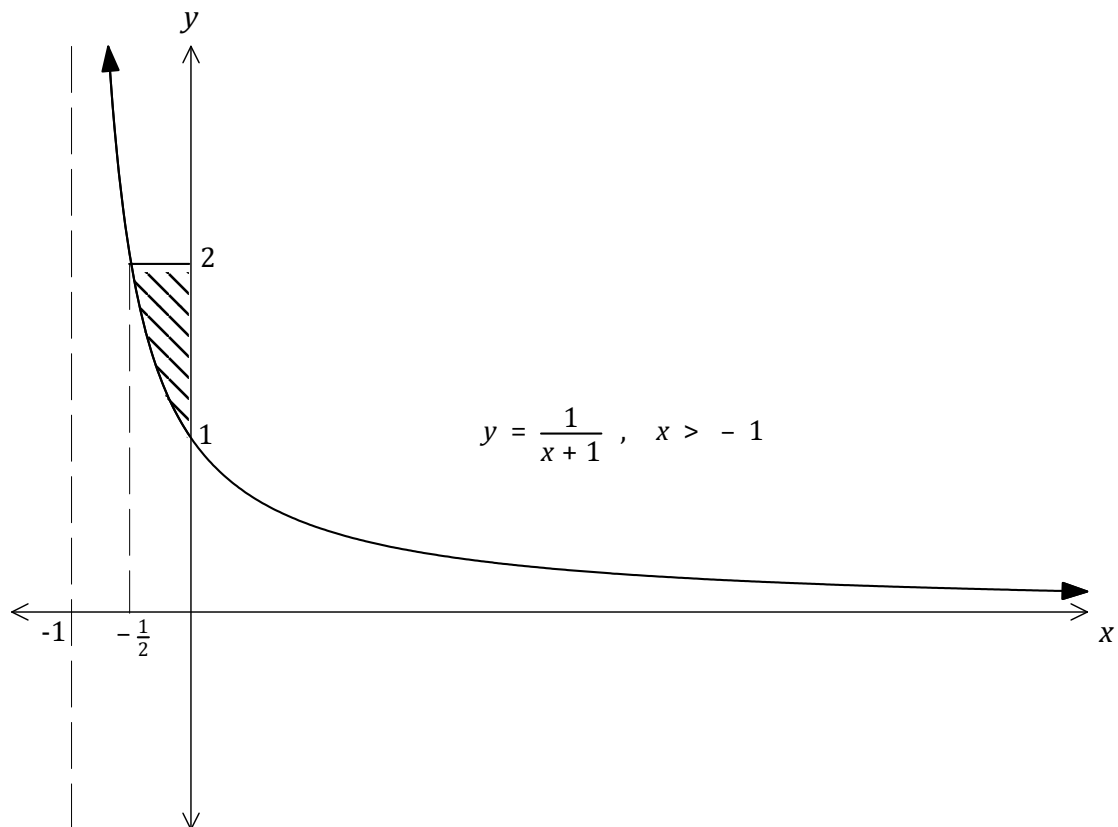
Let $P(x, 1 - x^2)$ be a point on this arc where $x > 0$

The rectangle PQRS is constructed as shown in the diagram.

- | | |
|---|----------|
| (i) Show that the area A of the rectangle is given by $A = 2x - 2x^3$ | 1 |
| (ii) Find the exact value of x in simplest form if PQRS is a square. | 2 |
| (iii) Prove that the maximum area of the rectangle is $\frac{4\sqrt{3}}{9}$ | 4 |
| (iv) Hence, or otherwise, find the minimum area enclosed between the parabolic arc and the rectangle. | 1 |

Question 8 continues on the next page

(b)



The shaded region bounded by the hyperbola $y = \frac{1}{x+1}$ and the y axis between $y = 1$ and $y = 2$ is revolved about the y axis.

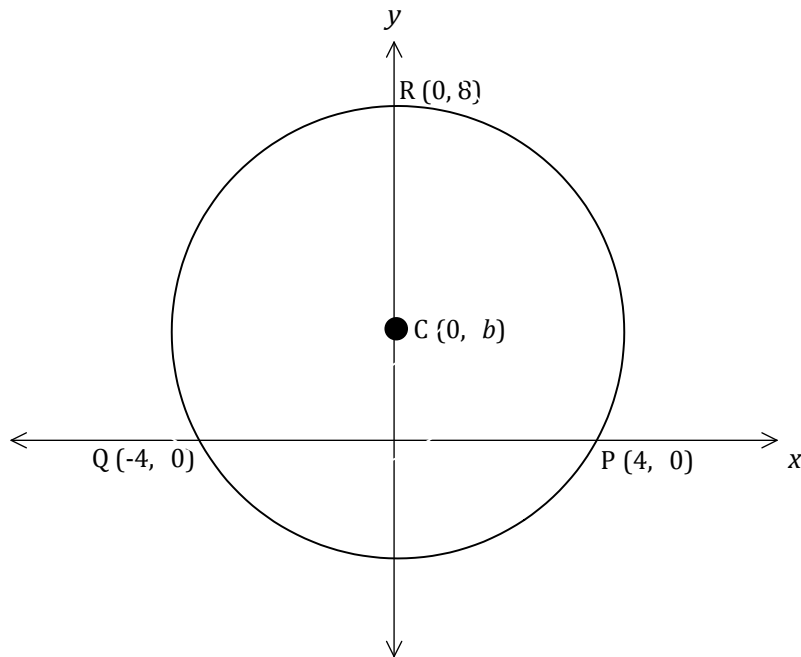
Find the volume of the solid of revolution.

4

End of Question 8

(a) Find the values for k for which the equation $x + \frac{1}{x} = 2k$ has no real solutions for x . **3**

(b)



The sketch shows the circle with centre $C(0, b)$ passing through the points $P(4, 0)$, $Q(-4, 0)$ and $R(0, 8)$.

(i) Find the radius of the circle. **2**

(ii) Hence, or otherwise, find the length of the minor arc PQ. Give your answer to 1 decimal place. **2**

Question 9 continues on the next page

-
- (c) A tank contains 2009 L of water. It is leaking so that after t hours the volume V litres of water is decreasing at a rate of R L/h where $R = 2t - 90$.

The rate formula for R applies until the tank is empty.

- | | | |
|-------|---|----------|
| (i) | Find the rate at which the volume is decreasing after 20 hours. | 1 |
| (ii) | Find the volume of water in the tank after 20 hours. | 2 |
| (iii) | When is the tank empty? | 2 |

End of Question 9

- (a) Sketch $y = \sec x$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$, showing any intercepts made with the coordinate axes. **1**

- (b) Sketch $y = \tan x$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$, and hence, or otherwise, evaluate

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \tan x \, dx$$
2

- (c) Let $f(x) = (1 + \tan x)^2$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$

- (i) Solve $f(x) = 0$ **1**

- (ii) Find any stationary points on $y = f(x)$ and determine their nature. **3**

- (iii) Sketch the graph of $y = f(x)$ **2**

- (iv) Find the area bounded by the curve and the x axis from $x = -\frac{\pi}{4}$ to $x = \frac{\pi}{4}$ **3**

End of Examination

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note: $\ln x = \log_e x, \quad x > 0$



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Mathematics

Question	Algebra and Number	Geometry	Functions	Trigonometry	Differential Calculus	Integral Calculus	Total
1	(a)(b)(c)(f) 8		(d) 2		(e) 2		/ 12
2				(a) 4	(b)(c) 8		/ 12
3		(b) 4	(a) 8				/ 12
4			(c)(d) 7			(a)(b) 5	/ 12
5			(a) 4	(b) 6		(c) 2	/ 12
6					12		/ 12
7		(b) 5	(a) 7				/ 12
8					(a) 8	(b) 4	/ 12
9	(a) 3		(b) 4		(c) 5		/ 12
10				(a) 1	(c)(i)-(iii) 6	(b)(c)(iv) 5	/ 12
Total	11	9	32	11	41	16	/120

Question 1

(a) 32.62

(b) $-8x < -16$

$\therefore x > 2$

(c) $\frac{1+x}{1+x} + x'' \frac{(1+x)}{1+x} = 1 + x''$

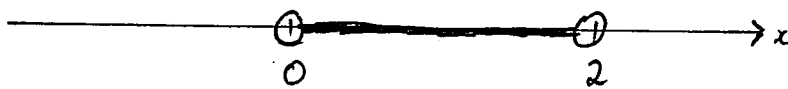
OR, of course, $\frac{1+x + x''(1+x)}{1+x} = \frac{(1+x)(1+x'')}{1+x} = 1+x''$

(d) $d = \frac{3 + 8 + 1}{\sqrt{6^2 + 8^2}} = \frac{12}{10} = 1.2$

(e) $f'(x) = \frac{2x}{10} = \frac{120}{10}$ for $x=60$

ie. $f'(60) = 12$

(f)



ie. $0 < x < 2$

Question 2

$$(a) (i) \text{ Area} = \frac{1}{2} \cdot 8 \cdot 5 \sin 60^\circ \text{ cm}^2 = 20 \cdot \frac{\sqrt{3}}{2} \text{ cm}^2 = 10\sqrt{3} \text{ cm}^2$$

$$(ii) BC^2 = 8^2 + 5^2 - 2 \cdot 8 \cdot 5 \cos 60^\circ$$

$$= 64 + 25 - 40 = 49$$

$$\therefore BC = 7 \text{ cm}$$

$$(b) (i) 3(1+2x)^2 \cdot 2 = 6(1+2x)^2$$

$$(ii) x \cdot \frac{1}{x} + \ln x \cdot 1 = 1 + \ln x$$

$$(c) (i) f'(1) = 1 + 2 - 3 = 0$$

$\therefore P(1, 3)$ is a stationary point

$$(ii) f''(x) = 2 - 6x$$

$$f''(1) = -4 < 0 \Rightarrow \cap \text{ ie. maximum turning point}$$

$$(iii) f(x) = x + x^2 - x^3 + C$$

$$\therefore f(1) = 1 + 1 - 1 + C = 3 \Rightarrow C = 2$$

$$\therefore f(x) = x + x^2 - x^3 + 2$$

Question 3

(a) (i) $M = \left(\frac{4}{2}, \frac{8}{2}\right) = (2, 4)$

(ii) Gradient of $AB = \frac{6}{12} = \frac{1}{2}$

$\therefore$ Gradient of $MP = -2$

$\therefore$ MP is $y - 4 = -2(x - 2)$

$\sim y - 4 = -2x + 4$

i.e. $2x + y = 8$

(iii) $P = (4, 0)$, $Q = (0, 8)$

(iv) The mid-point of $PQ = \left(\frac{4}{2}, \frac{8}{2}\right) = (2, 4) = M$

$\therefore$ diagonals AB , PQ bisect each other at right angles

$\Rightarrow APBQ$ is a rhombus

[ALTERNATIVES, OF COURSE]

(v) $AB = \sqrt{12^2 + 6^2} = 6\sqrt{5}$

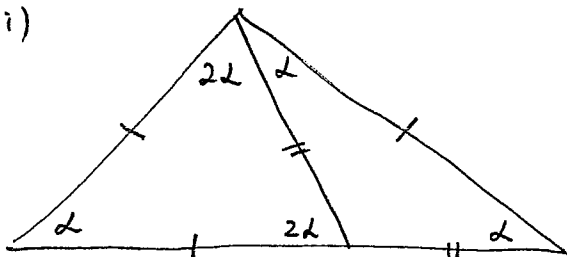
$PQ = \sqrt{4^2 + 8^2} = 4\sqrt{5}$

$\therefore$ Area = $\frac{1}{2} \cdot 6\sqrt{5} \cdot 4\sqrt{5} = 60$

(b) (i) $\angle CAP = \angle PCA = \alpha$, base $\angle$ s of isos $\triangle APC$

$\therefore \angle APB = \angle CAP + \angle PCA = 2\alpha$, ext. $\angle$ theorem in $\triangle APC$

(iii)



$\therefore 5\alpha = 180$

$\alpha = 36$

Question 4

$$(a) \quad I = 3 \int \frac{4x^3}{x^4+1} dx = 3 \ln(x^4+1)$$

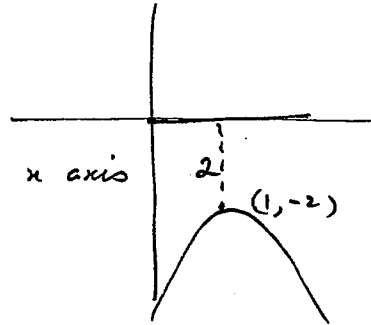
$$(b) \quad I \approx \frac{1}{6} \cdot 2 (1 + 5 + 4 \times 2) = \frac{14}{3}$$

$$(c) \quad (i) \quad V = (1, -2)$$

$$(ii) \quad a = 2 \quad \therefore$$

$\Rightarrow$ directrix is the x axis

$$\text{i.e. } y = 0$$



$$(d) \quad (i) \quad 2 + \beta = 4$$

$$2\beta = 2$$

$$(ii) \quad 3\alpha + 3\beta = 3(2 + \beta) = 12$$

$$3\alpha \cdot 3\beta = 9\alpha\beta = 18$$

$$\therefore \text{Equation is } x^2 - 12x + 18 = 0$$

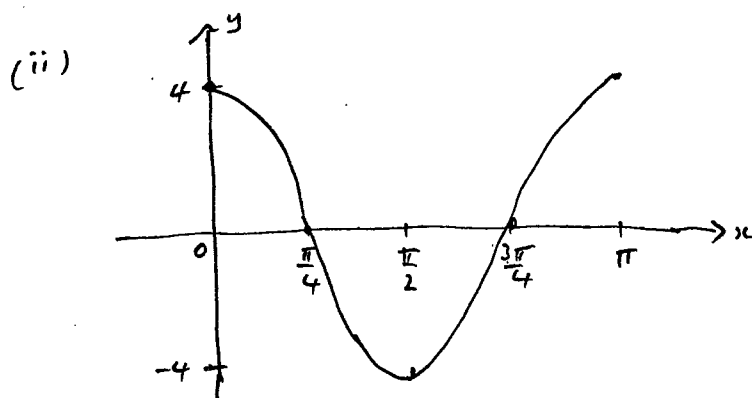
Question 5

(a) (i) $d = -9$

$$\therefore S_{14} = \frac{14}{2} (60 - 9 \times 13) = -399$$

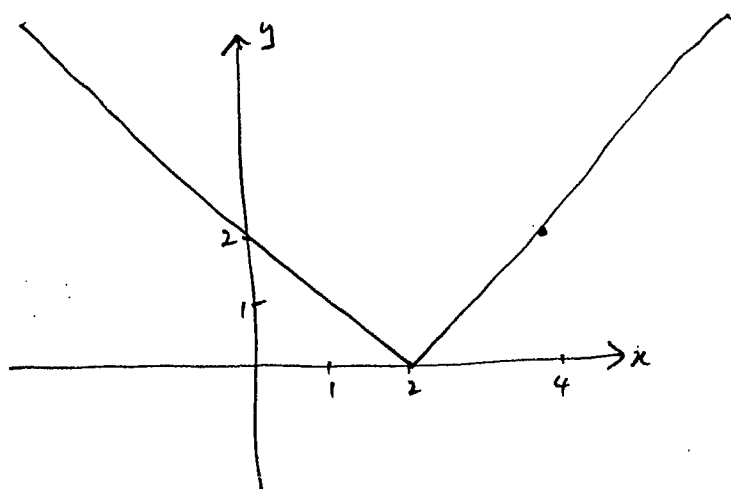
(ii) $r = \frac{21}{30} = \frac{7}{10} \quad \therefore S_{\infty} = \frac{30}{1 - \frac{7}{10}} = 100$

(b) (i) period = π



(iii) $A = \int_0^{\pi/4} 4 \cos 2x \, dx$
 $= \frac{4 [\sin 2x]}{2} \Big|_0^{\pi/4} = 2(1 - 0) = 2$

(c) (i)



(ii) $\int_0^4 |x - 2| \, dx = 2 \cdot \frac{1}{2} \cdot 2 \cdot 2 = 4$

Question 6

(a) (i) $t=0, x = 16 - 8 = 8$ (m)

(ii) $x = t + 16(t+1)^{-1} - 8$

$$\therefore \dot{x} = 1 - 16(t+1)^{-2} = 1 - \frac{16}{(t+1)^2}$$

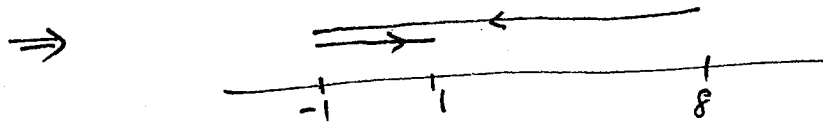
Stops when $\dot{x} = 0 \Rightarrow (t+1)^2 = 16$

$$\Rightarrow t+1 = 4 \text{ since } t \geq 0$$

i.e. stops at $t = 3$

(iii) $t = 3, x = 3 + 4 - 8 = -1$

$t = 7, x = 7 + 2 - 8 = 1$



distance travelled is 11 (m)

(iv) $\ddot{x} = 32(t+1)^{-3} = \frac{32}{(t+1)^3}$ has maximum value when $t = 0$

$$\Rightarrow \max \ddot{x} = 32 \text{ (m/s}^2\text{)}$$

(b) (i) If $P = Ae^{kt}$ then $\frac{dP}{dt} = Ae^{kt} \cdot k = k(Ae^{kt}) = kP$

(ii) $P = Ae^{kt}$ $t=0, P=4000$
 $t=12, P=5400$

$$\therefore P = 4000 e^{kt}$$

$$\therefore 5400 = 4000 e^{12k}$$

$$\Rightarrow e^{12k} = \frac{54}{40} = 1.35$$

$$\therefore 12k = \ln 1.35$$

$$k = \frac{1}{12} \ln 1.35 = 0.025, \text{ 2 sig figs}$$

(iii) at $t=0, \frac{dP}{dt} = 0.025 \times 4000 = 100$ people/yr

Question 7

$$(a) (i) \quad 6\% \text{ p.a.} = \frac{6}{12} \% \text{ p.month} = 0.005 \text{ p.month}$$

$$\therefore A_1 = P(1 + 0.005) - 232 = P(1.005) - 232$$

$$(ii) \quad A_2 = A_1(1.005) - 232 = P(1.005)^2 - 232 \times 1.005 - 232 \\ = P(1.005)^2 - 232(1 + 1.005)$$

$$(iii) \quad A_n = P(1.005)^n - 232(1 + 1.005 + 1.005^2 + \dots + 1.005^{n-1}) \\ = P(1.005)^n - 232 \frac{(1.005^n - 1)}{1.005 - 1}$$

$$= P(1.005)^n - 46400(1.005^n - 1)$$

$$(iv) \quad \text{Now } A_{60} = 0$$

$$\therefore P(1.005)^{60} = 46400(1.005^{60} - 1)$$

$$P = \frac{46400(1.005^{60} - 1)}{1.005^{60}} = \$12000, \text{ whole dollars}$$

$$(v) \quad \text{Owes } A_{12} = 12000(1.005)^{12} - 46400(1.005^{12} - 1) \\ = \$9878.28 \quad \text{or } \$9878 \text{ whole dollars}$$

$$(b) (i) \quad \angle A \text{ is common}$$

$$\angle B = \angle Q, \text{ given}$$

$$\therefore \triangle ABC \parallel \triangle APQ, \text{ equiangular}$$

$$(ii) \quad \therefore \frac{x}{12} = \frac{4}{x+2}$$

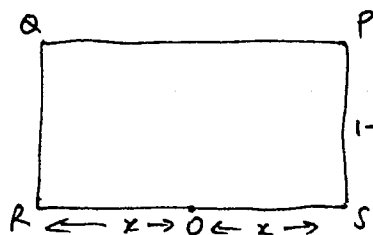
$$\Rightarrow x^2 + 2x = 48$$

$$x^2 + 2x - 48 = 0$$

$$(x + 8)(x - 6) = 0 \Rightarrow x = 6 \quad \text{since } x > 0$$

Question 8

(a) (i)



$$\therefore A = 2x(1-x^2) \\ = 2x - 2x^3$$

(ii) Here $2x = 1-x^2$

$$\text{or } x^2 + 2x - 1 = 0$$

$$\therefore x = \frac{-2 \pm \sqrt{4+4}}{2} \quad \text{since } x > 0$$

$$= \frac{-2 + 2\sqrt{2}}{2} = \sqrt{2} - 1$$

(iii) $\frac{dA}{dx} = 2 - 6x^2 = 0$ if $x^2 = \frac{1}{3}$ i.e. $x = \frac{1}{\sqrt{3}}$ since $x > 0$

$$\frac{d^2A}{dx^2} = -12x < 0 \quad \text{if } x = \frac{1}{\sqrt{3}} \Rightarrow \wedge \quad \text{i.e. maximum area when } x = \frac{1}{\sqrt{3}}$$

$$\therefore \max A = 2 \cdot \frac{1}{\sqrt{3}} \left(1 - \frac{1}{3}\right) = \frac{4}{3\sqrt{3}} = \frac{4\sqrt{3}}{9}$$

$$(iv) \text{ Min area} = \frac{1}{6} \cdot 2(4) - \frac{4\sqrt{3}}{9} = \frac{4}{3} - \frac{4\sqrt{3}}{9}$$

$$(b) V = \pi \int_1^2 x^2 dy \quad \text{where } y = \frac{1}{x+1} \quad \text{or } x+1 = \frac{1}{y} \\ \therefore x = \frac{1}{y} - 1$$

$$\therefore V = \pi \int_1^2 \left(\frac{1}{y} - 1\right)^2 dy$$

$$= \pi \int_1^2 \frac{1}{y^2} - \frac{2}{y} + 1 dy$$

$$= \pi \int_1^2 y^{-2} - \frac{2}{y} + 1 dy = \pi \left[-\frac{1}{y} - 2\ln y + y \right]_1^2$$

$$= \pi \left(-\frac{1}{2} - 2\ln 2 + 2 - (-1 - 0 + 1) \right)$$

$$= \pi \left(\frac{3}{2} - 2\ln 2 \right)$$

Question 9

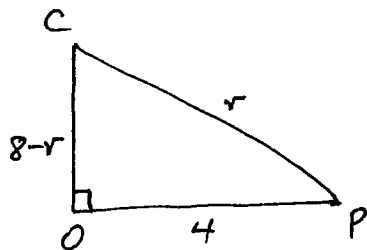
$$(a) \Rightarrow x^2 + 1 = 2kx$$

$$\therefore x^2 - 2kx + 1 = 0$$

$$\Delta = 4k^2 - 4 < 0 \text{ for no roots}$$

$$\Rightarrow k^2 < 1 \quad \text{i.e.} \quad -1 < k < 1$$

(b) (i)

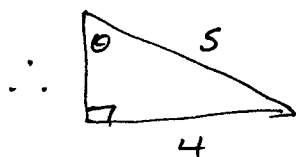


$$\therefore r^2 = (8-r)^2 + 16 = 64 - 16r + r^2 + 16$$

$$\therefore 16r = 80$$

$$r = 5$$

(ii)



$$\sin \theta = \frac{4}{5}, \quad \theta = 0.927 \dots$$

$$\therefore \angle PCQ = 2 \times 0.927 \dots$$

$$\therefore \text{arc PQ} = 5 \times 2 \times 0.927 \dots = 9.3, \text{ 1 d.p.}$$

(c) (i) $R = 2 \times 20 - 90 = -50$ i.e. decreasing at 50 L/h

$$(ii) \quad R = \frac{dV}{dt} = 2t - 90$$

$$\therefore V = t^2 - 90t + c$$

$$\therefore 2009 = 0 + c, \quad c = 2009$$

$$\therefore V = t^2 - 90t + 2009$$

$$\therefore t = 20, \quad V = 20^2 - 1800 + 2009 = 609 \text{ L}$$

$$\text{or } \frac{dV}{dt} = 2(t - 45)$$

$$\therefore V = (t - 45)^2 + c$$

$$\therefore 2009 = 45^2 + c, \quad c = 16$$

$$\therefore V = (t - 45)^2 + 16$$

$$t = 20, \quad V = 25^2 + 16 = 609 \text{ L}$$

(iii) Empty when $V = 0$

$$\Rightarrow t^2 - 90t + 2009 = 0$$

$$(t - 41)(t - 49) = 0$$

$$\Rightarrow t = 41 \quad \text{since } 41 < 49$$

i.e. empty after 41 hours

$$V = 0$$

$$\Rightarrow (t - 45)^2 = 16$$

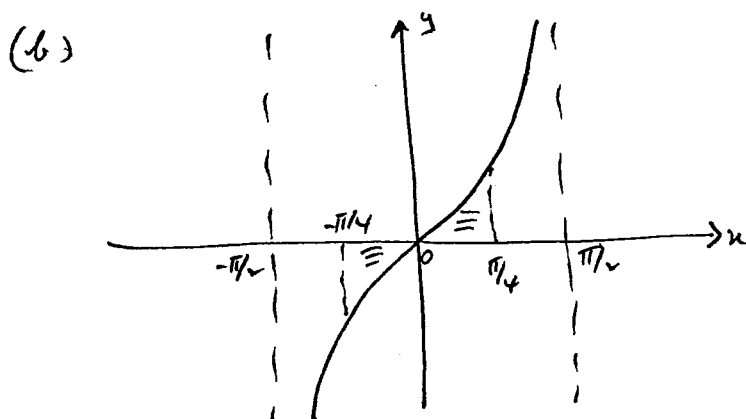
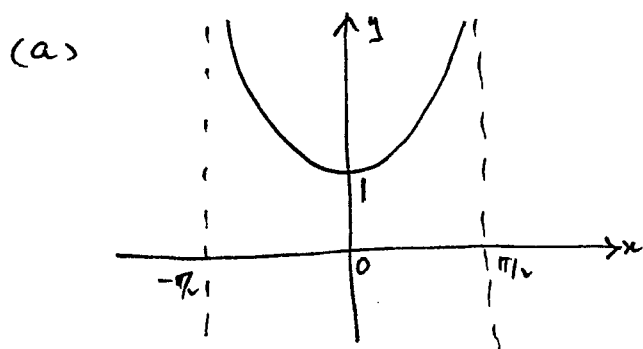
$$\therefore t - 45 = 4 \text{ or } -4$$

$$t = 49 \text{ or } 41$$

$$\Rightarrow t = 41 < 49$$

$\therefore \dots$

Question 10



From sketch,

$$\int_{-\pi/4}^{\pi/4} \tan x \, dx = 0$$

(c) (i) $f(x) = 0 \Rightarrow 1 + \tan x = 0$

i.e. $\tan x = -1 \Rightarrow x = -\frac{\pi}{4}$ since $-\frac{\pi}{2} < x < \frac{\pi}{2}$

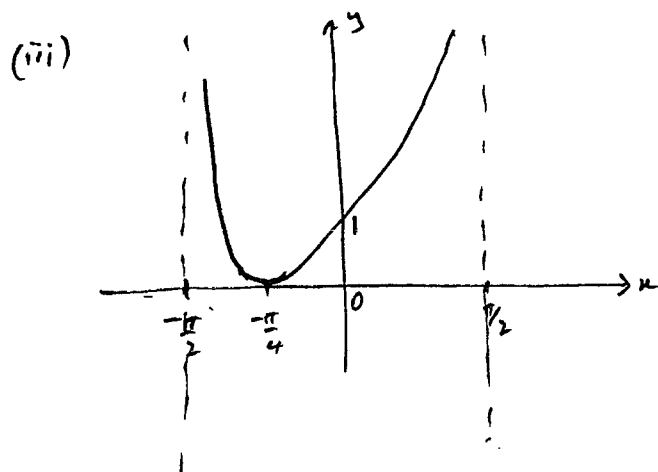
(ii) $f'(x) = 2(1 + \tan x) \sec^2 x = 0$ if $1 + \tan x = 0$

i.e. $x = -\frac{\pi}{4}$ from (i)

≈ -0.8

x	-0.9	-0.8	-0.7
$f'(x)$	< 0	0	> 0

$\Rightarrow$ min turning point $(-\frac{\pi}{4}, 0)$



$$(iv) \quad A = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} (1 + \tan x)^2 dx$$

$$= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} 1 + 2 \tan x + \tan^2 x dx$$

$$= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sec^2 x + 2 \tan x dx$$

$$= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sec^2 x dx \quad \text{since} \quad \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \tan x dx = 0 \quad \text{from (b)}$$

$$= \left[\tan x \right]_{-\frac{\pi}{4}}^{\frac{\pi}{4}} = 1 + 1 = 2$$

Qn 1 (a) 1 for correct 2 d.p.

Qn 2 (b) (i) Allow 1 for $3(1+2x)^2$
(ii) Allow 2 for $x \cdot \frac{1}{x} + \ln x \cdot 1$

(c) (ii) Allow 1 for $f''(1) = -4$

[Alternative solutions exist, of course]

(iii) Allow 1 for $f(x) = x + \tilde{x} - x^3 + c$

Qn 3 (a) (i) Allow 1 for $(\frac{4}{2}, \frac{8}{2})$

(ii) Allow 1 for $\text{grad PQ} = -2$

(b) Alternative solutions, of course

Allow 1 for distances AB, PQ

Qn 4 (a) Allow 1 for $3 \int \frac{4x^3}{x^4+1} dx$. No need for $+c$

(b) Allow a sensible decimal approximation

Allow 2 for $\frac{1}{6} \cdot 2 (\sqrt{1} + \sqrt{25} + 4\sqrt{4})$

(c) (ii) Allow 1 for $a = 2$

(d) (ii) Allow 1 for $3\alpha + 3\beta = 12$ and $3\alpha \cdot 3\beta = 18$

Qn 5 (a) (i) Allow 1 for $d = -9$

(ii) Allow 1 for $r = \frac{21}{30}$

Don't allow 2 for $S_{\infty} = \frac{30}{1 - \frac{7}{10}}$

(b) (iii) Allow 1 for $\int_0^{\frac{\pi}{4}} 4 \cos 2u \, du$

1 for $4 \left[\frac{\sin 2u}{2} \right]_0^{\pi/4}$

Qn 6 (a) Ignore units

(ii) Allow 1 for $\ddot{x} = 1 - 16(t+1)^{-2} = 0$

(iii) Allow 1 for $t = 3, x = -1$ and $t = 7, x = 1$

(iv) Allow 1 for $\ddot{x} = 32(t+1)^{-3}$

(b) (ii) Allow 1 for $A = 4000$ quoted or otherwise

Allow 0 for $5400 = 4000 e^{12k}$

Allow 1 for $12k = \ln\left(\frac{54}{40}\right)$

Allow 1 for 2 significant figures

Question 7

(a) (ii) Need to indicate $A_2 = A_1(1.005) - 232$

(iii) Allow 1 for $A_n = P(1.005)^n - 232(1 + 1.005 + \dots + 1.005^{n-1})$

(iv) Allow 2 for $\$12000.33$

Allow 1 for $A_{60} = 0$

(b) (ii) Allow 1 for $\frac{x}{12} = \frac{4}{x+2}$ or equivalent

Allow 1 for $x^2 + 2x - 48 = 0$

Allow 3 if they determine $x=6$ by "trial" with clear reasons

Qn 8 (a) (i) Allow 1 for $A = 2x(1-x^2)$

(ii) Allow 1 for $x = \frac{-2 \pm \sqrt{4+4}}{2}$

(iii) Allow 1 for $\frac{dA}{dx} = 2 - 6x^2 = 0$

IGNORE UNITS

Allow 1 for $x = \frac{1}{\sqrt{3}}$ or $\frac{\sqrt{3}}{3}$, of course

Deduct 1 if check for maximum not shown

(iv) Allow 1 for correct use of Simpson's rule

or Min area = $\int_{-1}^1 (1-x^2) dx = \frac{4\sqrt{3}}{9}$ or equivalent

(b) Allow 1 for $V = \pi \int_1^2 x^2 dy$

Allow 1 for $x = \frac{1}{y} - 1$ or equivalent

Allow 1 for $V = \pi \int_1^2 \left(\frac{1}{y^2} - \frac{2}{y} + 1 \right) dy$

Don't deduct marks for decimal approximation $[0.3572\dots]$

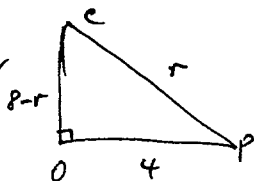
IGNORE UNITS

Qn 9

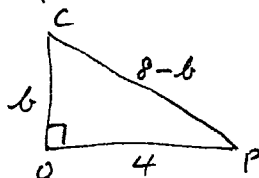
(a) Allow 1 for $x^2 - 2kx + 1 = 0$

Allow 1 for indicating $\Delta < 0$

(b) (i) Allow 1 for



or equivalent



(ii) Allow 1 for $\angle PCQ = 1.854 \dots$

(most will use cos rule)

Allow 2 for arc PQ = $5 \times 1.854 \dots$

i.e. ignore 1 d.p. instruction

(c) (i) Allow 1 for -50

(ii) Allow 1 for $V = t^2 - 90t + c$ or $\frac{2(t-45)^2}{2} + c$

(iii) Allow 1 for $t = 41, 49$

Qn 10

(c) (ii) Allow 1 for $f'(x) = 2(1 + \tan x) \sec x$

Allow 1 for $\Rightarrow 1 + \tan x = 0$

Need to do check of some sort to show min

(iii) Don't deduct 1 if y intercept not shown

(iv) Allow 1 for $A = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} 1 + 2 \tan x + \tan^2 x \, dx$

Allow 1 for $A = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sec^2 x + 2 \tan x \, dx$